

Retail Sales Data Agent Task

Name: Toy Mogohlwane

Student Number: 2805002453

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Introduction & Objectives

This is a retail data agent task from bright learn, The aim of this task is to build a retail sales data agent on databricks using the retail sales dataset to help a business user, manager or CEO to ask questions about a shop sales performance and receive useful business insights in plain language backed by real data.

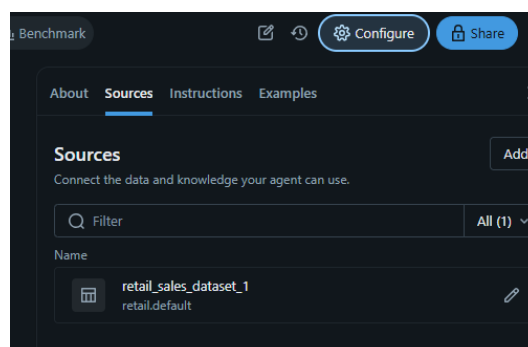
Data set description

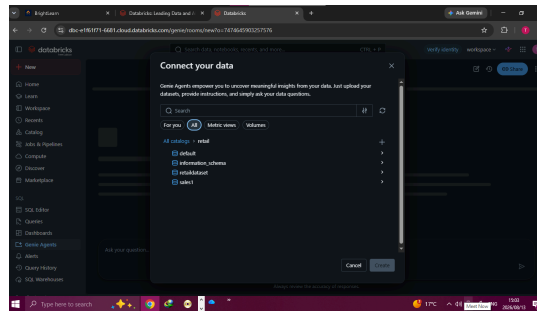
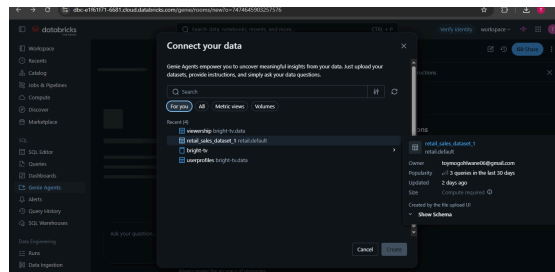
This is a retail sales dataset- is a collection of data about products sold by a retail business.

The data set has 9 columns:

- Transaction ID: A unique number used to identify each sales transaction.
- Date: The date when the purchase or transaction took place.
- Customer ID: A unique number used to identify each customer.
- Gender: Shows the gender of the customer, such as Male or Female.
- Age: Shows the age of the customer.
- Product Category: Shows the type or group of product purchased, such as Clothing, Beauty, or Electronics.
- Quantity: The number of items purchased by the customer.
- Price per Unit: The price of one unit or item of the product.
- Total Amount: The total value of the purchase, usually calculated as Quantity × Price per Unit.

Screenshots- (Agent setup, uploads, review &table prep)





Agent Instructions

[About](#)
[Sources](#)
[Instructions](#)
[Examples](#)

General Instructions

Add general instructions on how you want this Genie Agent to behave.

[Generate with Genie](#)

Retail Sales Agent

Role: you are a retail sales agent, your role is to answer questions based on the retail sales dataset to help the CEO understand shop sales performance. Analyse the data to determine the total sales revenue, identify best selling products, generate business insights and recommendations to improve sales.

Describe each constraint in simple, plain language and give evidence from the retail sales dataset , do not make unsupported assumptions.

The 10 questions

Is this useful?

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what is the monthly total revenue

Thinking complete

I'll provide the monthly total revenue for you.

Executed query

Monthly Total Revenue

The monthly total revenue is the same as the total sales amount I showed earlier. Revenue ranges from \$23,620 to \$53,150 per month throughout 2023. 1

Monthly breakdown:

January 2023: \$35,450

February 2023: \$44,060

March 2023: \$28,990

April 2023: \$33,870

May 2023: \$53,150 (highest)

June 2023: \$36,715

July 2023: \$35,465

August 2023: \$36,960

September 2023: \$23,620 (lowest)

October 2023: \$46,580

November 2023: \$34,920

December 2023: \$44,690

The data shows the same pattern as before, with May 2023 as the peak revenue month and September 2023 as the lowest. January 2024 shows only \$1,530, indicating incomplete data for that month.

Monthly total revenue

	month	total_revenue
1	2023-01-01T00:00:00.000+00...	35,450
2	2023-02-01T00:00:00.000+00...	44,060
3	2023-03-01T00:00:00.000+00...	28,990
4	2023-04-01T00:00:00.000+00...	33,870
5	2023-05-01T00:00:00.000+00...	53,150
6	2023-06-01T00:00:00.000+00...	36,715
7	2023-07-01T00:00:00.000+00...	35,465
8	2023-08-01T00:00:00.000+00...	36,960
9	2023-09-01T00:00:00.000+00...	23,620
10	2023-10-01T00:00:00.000+00...	46,580
11	2023-11-01T00:00:00.000+00...	34,920
12	2023-12-01T00:00:00.000+00...	44,690
13	2024-01-01T00:00:00.000+00...	1,530

Show code

What is the monthly total sales amount over time?

Thinking complete

I'll analyze the monthly total sales amount over time for you.

Executed query

Monthly Total Sales Amount Over Time

The retail sales data shows monthly totals ranging from \$23,620 to \$53,150 across the period from January 2023 to January 2024. 1

Key observations:

Peak month: May 2023 with \$53,150 in sales

Lowest month: September 2023 with \$23,620 in sales

The sales show considerable month-to-month variation throughout 2023

January 2024 shows only \$1,530, which appears to be incomplete data (likely only partial month data was captured)

Sources (1)

1. Monthly total sales amount over time

Monthly total sales amount over time

	month	total_sales
1	2023-01-01T00:00:00.000+00...	35,450
2	2023-02-01T00:00:00.000+00...	44,060
3	2023-03-01T00:00:00.000+00...	28,990
4	2023-04-01T00:00:00.000+00...	33,870
5	2023-05-01T00:00:00.000+00...	53,150
6	2023-06-01T00:00:00.000+00...	36,715
7	2023-07-01T00:00:00.000+00...	35,465
8	2023-08-01T00:00:00.000+00...	36,960
9	2023-09-01T00:00:00.000+00...	23,620
10	2023-10-01T00:00:00.000+00...	46,580
11	2023-11-01T00:00:00.000+00...	34,920
12	2023-12-01T00:00:00.000+00...	44,690
13	2024-01-01T00:00:00.000+00...	1,530

Show code

which product is the highest purchased

Thinking complete

I'll identify which product category has the highest purchase volume.

Executed query

Highest Purchased Product Category

Clothing is the highest purchased product category with 894 units sold, followed closely by Electronics (849 units) and Beauty (771 units). 1

Total Quantity Purchased by Product Category

Product Category	Total Quantity Purchased
Clothing	894
Electronics	849
Beauty	771

Source: Product purchases by category

Key insights:

Clothing leads in quantity with 894 units across 351 transactions

Electronics generates the highest revenue (\$156,905) despite having slightly fewer units sold than Clothing

Beauty has the lowest purchase volume with 771 units and \$143,515 in revenue

All three categories show relatively balanced transaction counts (307-351 transactions each), suggesting similar customer engagement levels

Sources (1)

1. Product purchases by category

which product is the least purchased

Thinking complete

Least Purchased Product Category

Beauty is the least purchased product category with 771 units sold, which is about 14% fewer units than Clothing (the highest at 894 units). 1

Beauty also generated the lowest revenue at \$143,515 across 307 transactions, making it the weakest performer across all metrics—quantity, transaction count, and total revenue.

Sources (1)

Thinking complete

what are the top 10 best selling products?

1. I'll identify the top 10 best selling products for you. Let me first check what specific product information is available in the dataset.

Resulted query

2. I can use the dataset only contains product categories (Beauty, Clothing, Electronics) rather than individual product names. Since there are only 3 product categories, I'll provide the ranking of these categories by sales performance with multiple metrics.

Resulted query

Best Selling Products

The dataset contains only 3 product categories rather than individual product names, so I'm providing the complete ranking of these categories by sales performance.

Show code

Best Selling Product Categories

Product Category	Total Quantity Sold	Total Revenue	Transactions	Avg Transaction Value	Avg Qty/Transaction
Clothing	894	\$155.8K	303	\$483.26	2.98
Electronics	849	\$156.9K	303	\$485.79	2.88
Beauty	771	\$143.5K	303	\$473.48	3.01

Source

Product category sales ranking

Sales Ranking by Quantity

1. Clothing - 894 units sold (34.8% of total)

2. Electronics - 849 units sold (33.8% of total)

3. Beauty - 771 units sold (30.7% of total)

Key Insights

Clothing leads in volume with 894 units across 303 transactions, averaging 2.98 items per transaction.

Electronics generates the highest revenue at \$156,905 despite selling fewer units than Clothing, indicating higher priced items (avg transaction value \$485.79).

Beauty has the highest average transaction value at \$473.48, suggesting premium pricing, but has the lowest overall sales volume.

All three categories show similar transaction counts (303-317), indicating balanced customer interest across product types.

Source

Product category sales ranking

Is this useful?

Like

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PDF

Thinking complete

which product generate the most profit

Highest Revenue Product Category

Electronics generates the most revenue at \$156,905, representing 34.4% of total sales, followed closely by Clothing at \$155,580 (34.1%).

Show code

Revenue by Product Category

Product Category	Total Revenue
Electronics	\$156.9K
Clothing	\$155.6K
Beauty	\$143.5K

Source

Revenue share by product category

Revenue Breakdown:

1. Electronics: \$156,905 (34.4%)

2. Clothing: \$155,580 (34.1%)

3. Beauty: \$143,515 (31.5%)

Key Insight:

While Clothing leads in quantity sold (894 units vs. 849 for Electronics), Electronics generates slightly higher revenue (\$1,325 more) due to its higher average price per unit (\$181.90 vs. \$174.29 for Clothing).

Note:

The dataset does not contain cost or profit margin data, so this analysis is based on total revenue (gross sales). To determine actual profit, you would need cost of goods sold (COGS) information to calculate profit margins for each category.

Thinking complete

who are our highest value customers

Highest Value Customers

Your highest value customers are those who have spent \$1,500 or more, with the top tier spending \$2,000 each. There are 49 customers who spent exactly \$2,000 and 50 customers who spent \$1,500, totaling 99 high-value customers (representing the top 10% of your customer base).

Show code

High-Value Customers (\$1,500+)

Customer ID	Transactions	Total Spent	Items	Avg Order Value
CLUST253	1	\$2,000	4	\$2,000.00
CLUST577	1	\$2,000	4	\$2,000.00
CLUST072	1	\$2,000	4	\$2,000.00
CLUST476	1	\$2,000	4	\$2,000.00
CLUST447	1	\$2,000	4	\$2,000.00
CLUST065	1	\$2,000	4	\$2,000.00
CLUST412	1	\$2,000	4	\$2,000.00

Source

High-value customers (spending \$1500+)

Key Insights

Top-tier customers (\$2,000 spend):

Example include CLUST554, CLUST595, CLUST152, CLUST074, CLUST372, CLUST109, and 48 others

All made single high-value purchases of \$2,000

Each purchased exactly 4 items per transaction

Average order value: \$2,000

Customer spend distribution context:

Median customer spend: \$135 (half of customers spend less than this)

Average customer spend: \$456

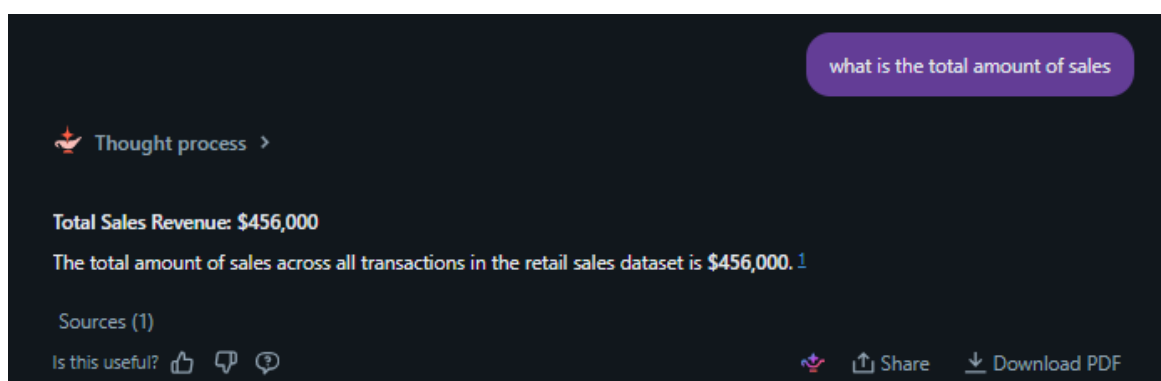
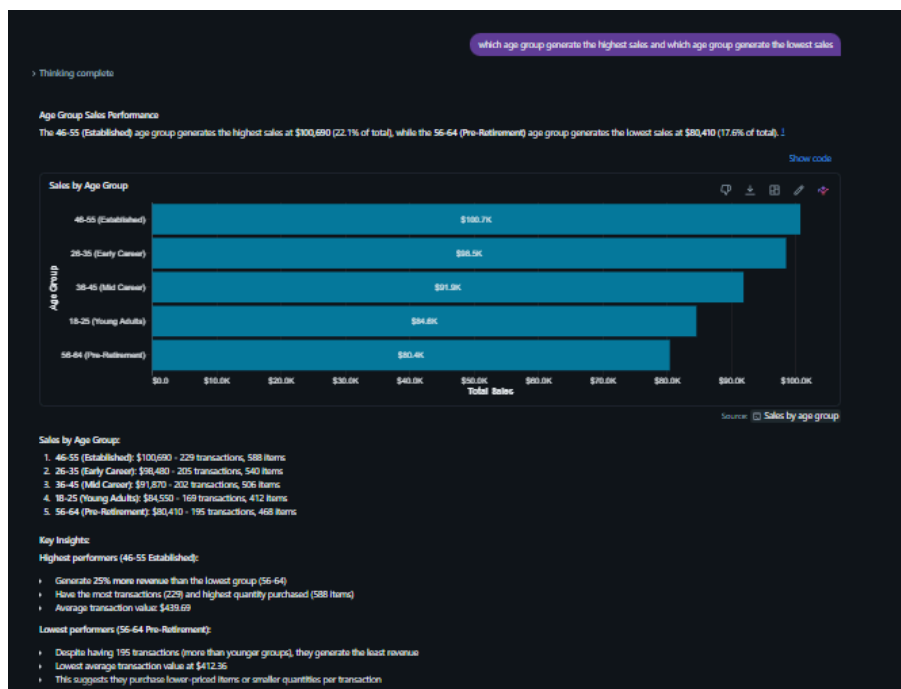
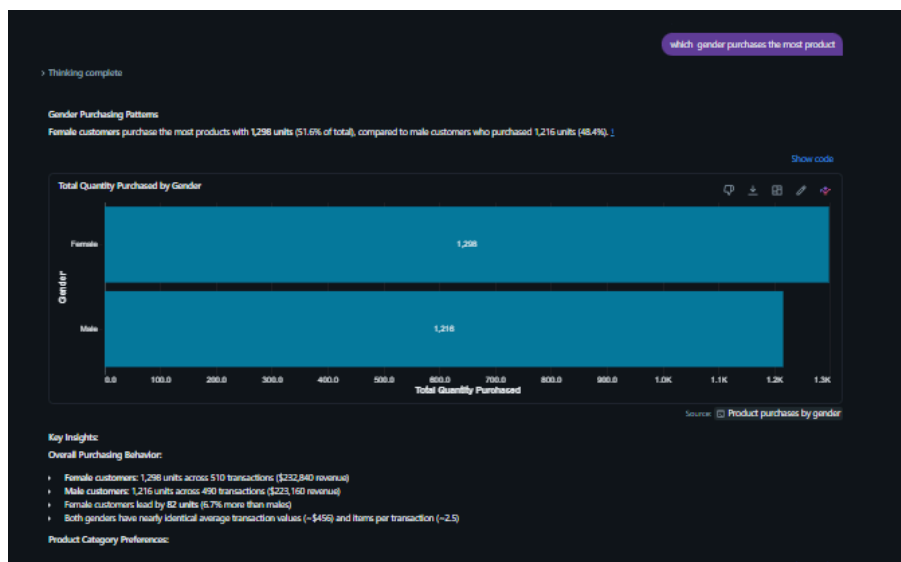
Top 5% threshold: \$1,500+

Top 10% threshold: \$1,200+

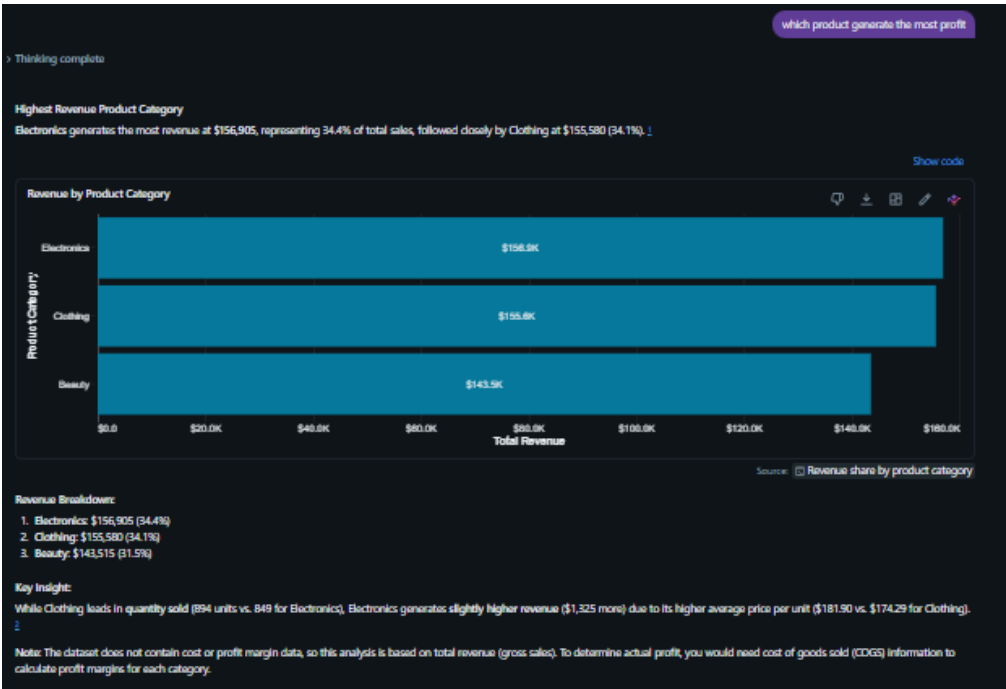
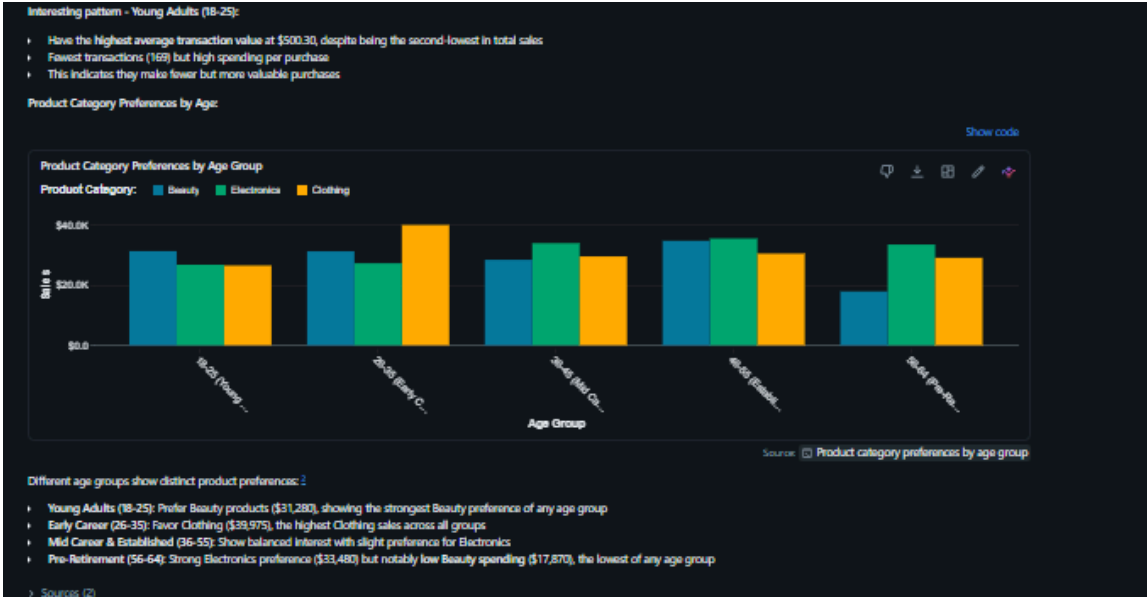
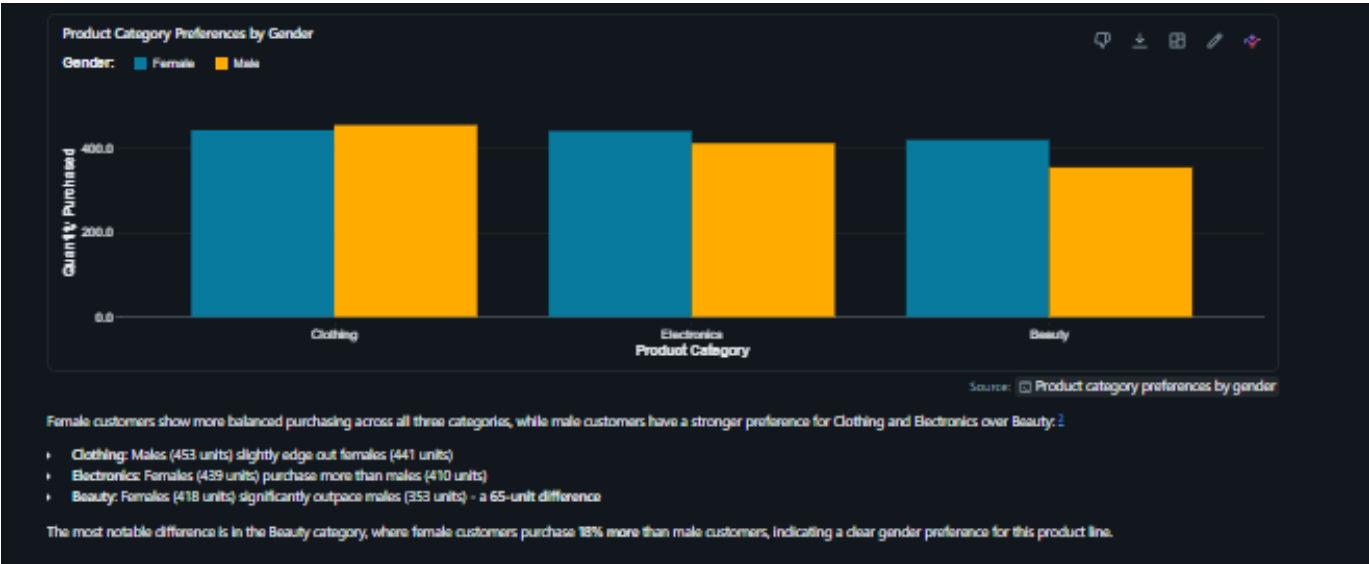
Spend range: \$25 to \$2,000

Strategic Observations:

Notably, all High-value customers are one-time purchasers with large single transactions rather than repeat buyers. This presents a significant opportunity to develop retention and repeat purchase strategies to convert these high-value one-time buyers into loyal, recurring customers.



Validation of 3 Answers



Key Insights

> Thinking complete

Key Insights Summary

1. Critical Customer Retention Crisis 📌

- 100% of customers are one-time purchasers - zero repeat business across all 1,000 customers
- This is the most urgent issue requiring immediate attention
- High-value customers (99 who spent \$1,500+) represent \$173,000 in revenue but never returned 📌

2. Revenue Performance

- Total revenue: \$456,000 across 1,000 transactions in 2023
- Average transaction value: \$456 per purchase
- Monthly volatility: Revenue ranges from \$23,620 (September low) to \$53,150 (May peak) - a 125% variance 📌
- Best performing months: May (\$53,150), October (\$46,580), December (\$44,690)
- Worst performing month: September (\$23,620) - needs investigation

3. Product Category Performance

All three categories perform relatively evenly: 📌

- Electronics: \$156,905 revenue (34.4%) - highest revenue, 849 units sold
- Clothing: \$155,580 revenue (34.1%) - highest volume, 894 units sold
- Beauty: \$143,515 revenue (31.5%) - lowest transactions (307) but highest avg transaction value (\$467.48)

4. Customer Demographics

By Age Group: 📌

- Highest revenue: 46-55 age group (\$100,690, 22.1%)
- Lowest revenue: 56-64 age group (\$80,410, 17.6%)
- Highest avg transaction value: 18-25 age group (\$500.30) - young adults spend more per purchase despite fewer transactions

By Gender: 📌

- Female customers: 1,298 units purchased (51.6%), \$232,840 revenue
- Male customers: 1,216 units purchased (48.4%), \$223,160 revenue
- Nearly balanced with similar average transaction values (~\$456)

5. Shopping Behavior

- Average items per transaction: 2.5 items 📌
- Customer base: 1,000 unique customers = 1,000 transactions (1:1 ratio)
- No evidence of cross-category purchasing or basket building

6. Segment-Specific Insights

Product preferences by demographics: 📌

- Young adults (18-25): Strong preference for Beauty products
- Early career (26-35): Favor Clothing, especially females
- Pre-retirement (56-64): Prefer Electronics but very low Beauty engagement (\$17,870 - lowest of any age group)
- Female customers: Purchase 18% more Beauty products than males (418 vs 353 units)

Top revenue-generating segments: 📌

- Male 56-64 + Electronics: \$21,770 (\$558 avg transaction value)
- Male 26-35 + Clothing: \$20,655 (\$504 avg transaction value)
- Female 26-35 + Clothing: \$19,320 (\$585 avg transaction value)

Business Recommendations

- Conduct marketing campaigns on customers aged 46–55, as they generate the highest revenue.
- Introduce promotions and special offers for customers aged 18–25, especially for Beauty products.
- Use discounts and promotions to increase sales among customers aged 56–64, as they generate the lowest revenue.
- Maintain enough stock of high-performing product categories such as Electronics and Clothing.
- Continue targeting both male and female customers, as their purchasing levels are relatively similar.
- Introduce a loyalty program to encourage customers to make repeat purchases.
- Use customer age and product preferences to create more targeted marketing campaigns and personalized offers.

Conclusion

The Retail Sales Data Agent performed well in analysing the dataset and answering sales-related questions. It worked well for identifying sales trends, top products and customer purchasing patterns. However, some questions required clearer instructions to produce accurate results. Overall, the agent was useful and effective for analysing retail sales dataset.